

Lectures 1–16 on Curves on an Algebraic Surface

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Abstract

These notes give a sustained modern treatment of the first sixteen lectures of Mumford's *Lectures on Curves on an Algebraic Surface*. The original table of contents for Lectures 1–16 runs from “Raw Material on Curves on Surfaces” to “The Method of Chow Schemes.” The present text keeps that architecture but rewrites the arguments in contemporary scheme-theoretic language, expands the proofs, and isolates the main deformation-theoretic and cohomological mechanisms. In particular, we make systematic use of: functors of points and Yoneda's lemma, Čech and Leray spectral sequences, Castelnuovo–Mumford regularity, flattening stratifications, the Hilbert functor, the universal family on the Hilbert scheme, and the Hilbert–Chow morphism. The outcome is not a line-by-line transcription of the 1966 text, but a mathematically faithful deepening of its first sixteen lectures, organized so that the main existence theorems for families of curves on a fixed smooth projective surface emerge from a single coherent framework.

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Standing assumptions and global strategy

Throughout k denotes an algebraically closed field unless a different hypothesis is explicitly imposed. A *surface* means a smooth projective k -scheme F of pure dimension 2. By a *curve on F* we mean either a purely one-dimensional closed subscheme $C \subset F$ or, when convenient, the associated effective divisor. Since F is regular, every codimension-one integral closed subscheme is locally principal; thus Weil and Cartier divisors coincide on F .

The leitmotif of Lectures 1–16 is that the geometry of families of curves on F can be controlled by three kinds of linear data:

- (i) cohomology of invertible sheaves on F and on the curves,

- (ii) tangent and obstruction spaces for deformations,
- (iii) numerical invariants encoded by Hilbert polynomials.

The classical text reaches the parameter spaces by Chow coordinates and flattening; modern treatments often pass through the Hilbert functor first and recover Chow spaces afterwards. We shall do both: the existence statements are proved with modern representability, while the classical Chow method is reconstructed in the final section.

1. Lecture 1: raw material on curves on surfaces and the problems suggested

1.1. Divisors, curves, and equivalence relations

Let F be a smooth projective surface. For each integral curve $\Gamma \subset F$ we write $[\Gamma]$ for the corresponding prime divisor. Every divisor is a finite sum

$$D = \sum_i n_i \Gamma_i, \quad n_i \in \mathbf{Z},$$

and every effective divisor determines a closed subscheme through its invertible ideal sheaf $\mathcal{O}_F(-D) \subset \mathcal{O}_F$.

Definition 1.1. Let D, D' be divisors on F .

- (a) They are linearly equivalent, written $D \sim D'$, if $D - D' = \text{div}(f)$ for some $f \in k(F)^\times$.
- (b) They are algebraically equivalent if they occur as fibres of a divisor on $F \times T$ over two k -points of a connected parameter scheme T .
- (c) They are numerically equivalent if $D \cdot E = D' \cdot E$ for every divisor E on F .

The resulting quotient groups are

$$\text{Pic}(F) = \frac{\{\text{Cartier divisors}\}}{\text{linear equivalence}}, \quad \text{NS}(F) = \frac{\text{Pic}(F)}{\text{Pic}^0(F)}, \quad \text{Num}(F) = \frac{\text{Pic}(F)}{\equiv_{\text{num}}}.$$

For a smooth projective surface one knows that $\text{NS}(F)$ is finitely generated and that the quotient $\text{NS}(F) \twoheadrightarrow \text{Num}(F)$ has finite kernel.

1.2. Intersection theory on a smooth surface

If D and E are divisors without common components, their intersection product is

$$D \cdot E = \sum_{x \in D \cap E} i_x(D, E),$$

where $i_x(D, E) = \dim_k \mathcal{O}_{F,x} / (f, g)$ for local equations f, g of D, E in the regular local ring $\mathcal{O}_{F,x}$. Bilinearity then extends the product to all divisors.

Proposition 1.2. The intersection pairing

$$\text{Div}(F) \times \text{Div}(F) \longrightarrow \mathbf{Z}$$

descends to a bilinear pairing on $\text{Num}(F)$.

Proof. Bilinearity is clear from the definition in regular local rings. If $D = \text{div}(f)$ is principal and E is prime, then one computes locally on the normalization of E that the sum of valuations of $f|_E$ is zero; hence $D \cdot E = 0$. Therefore the pairing factors through linear equivalence in the first variable, and by symmetry also in the second. By definition of numerical equivalence it factors through $\text{Num}(F)$. $\square$

1.3. Hilbert polynomial of a curve on a surface

Fix a very ample divisor H on F . If $C \subset F$ is a purely one-dimensional closed subscheme, define

$$P_C(m) = \chi(C, \mathcal{O}_C(mH)).$$

Because C has dimension one, P_C is linear.

Proposition 1.3. *Let $C \subset F$ be a purely one-dimensional subscheme. Then*

$$P_C(m) = (H \cdot C)m + \chi(\mathcal{O}_C).$$

If C is integral, writing $p_a(C) = 1 - \chi(\mathcal{O}_C)$, one gets

$$P_C(m) = (H \cdot C)m + 1 - p_a(C).$$

Proof. Use the exact sequence

$$0 \rightarrow \mathcal{O}_C((m-1)H) \rightarrow \mathcal{O}_C(mH) \rightarrow \mathcal{O}_{H \cap C}(mH) \rightarrow 0$$

for a general hyperplane section H . Since $H \cap C$ is a finite scheme of length $H \cdot C$, one has

$$\chi(\mathcal{O}_C(mH)) - \chi(\mathcal{O}_C((m-1)H)) = H \cdot C.$$

Hence $P_C(m)$ is linear of slope $H \cdot C$. Evaluating at $m = 0$ gives the constant term $\chi(\mathcal{O}_C)$. $\square$

1.4. The moduli problems already visible in Lecture 1

Mumford isolates several questions which, in modern language, can be phrased as follows.

- (Q1) For fixed numerical data (degree, divisor class, arithmetic genus, Hilbert polynomial), does there exist a parameter space of curves on F ?
- (Q2) Can one recover the dimension of those parameter spaces from linear data such as $h^0(C, \mathcal{N}_{C/F})$ and $h^1(F, \mathcal{O}_F)$?
- (Q3) How do line bundles and their complete linear systems control the irreducible components of the parameter spaces?
- (Q4) Which divisor classes are “good” in the sense that they are effective, generated by global sections, very ample, or represented by reduced curves?

The rest of the first sixteen lectures build the mechanism for answering these questions.

2. Lecture 2: the fundamental existence problem and two analytic proofs

From this point on assume $k = \mathbf{C}$ when explicitly stated.

2.1. The exponential-sequence proof of $\dim \text{Pic}^0(F) = h^1(F, \mathcal{O}_F)$

Let F be a smooth projective complex surface. The analytic exponential sequence

$$0 \rightarrow \mathbf{Z} \rightarrow \mathcal{O}_F \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}_F^\times \rightarrow 0$$

yields the long exact sequence

$$H^1(F, \mathbf{Z}) \rightarrow H^1(F, \mathcal{O}_F) \rightarrow H^1(F, \mathcal{O}_F^\times) \xrightarrow{c_1} H^2(F, \mathbf{Z}).$$

The kernel of c_1 is by definition $\text{Pic}^0(F)$.

Theorem 2.1. *For a smooth projective complex surface F there is a canonical isomorphism of complex Lie groups*

$$\mathrm{Pic}^0(F) \cong H^1(F, \mathcal{O}_F) / H^1(F, \mathbf{Z}),$$

where the lattice $H^1(F, \mathbf{Z})$ is viewed inside $H^1(F, \mathcal{O}_F)$ via the map induced by the inclusion $\mathbf{Z} \hookrightarrow \mathcal{O}_F$. In particular,

$$\dim \mathrm{Pic}^0(F) = h^1(F, \mathcal{O}_F) =: q(F).$$

Proof. The exponential sequence shows that $\mathrm{Pic}^0(F) = \ker(c_1)$, and exactness identifies this kernel with the cokernel of the map $H^1(F, \mathbf{Z}) \rightarrow H^1(F, \mathcal{O}_F)$. Because $H^1(F, \mathbf{Z})$ is a finitely generated abelian group and its image spans $H^1(F, \mathcal{O}_F)$ over $\mathbf{R}$, the quotient is a compact complex torus of complex dimension $h^1(F, \mathcal{O}_F)$. $\square$

2.2. A second analytic proof via Albanese duality

The Hodge decomposition gives

$$H^1(F, \mathbf{C}) \cong H^{1,0}(F) \oplus H^{0,1}(F), \quad h^{1,0}(F) = h^{0,1}(F) = q(F).$$

The Albanese torus is

$$\mathrm{Alb}(F) = H^0(F, \Omega_F^1)^\vee / H_1(F, \mathbf{Z}),$$

and its dual torus is naturally

$$\mathrm{Alb}(F)^\vee \cong H^{0,1}(F) / H^1(F, \mathbf{Z}).$$

Theorem 2.2. *For a smooth projective complex surface F , the connected Picard variety is canonically dual to the Albanese variety:*

$$\mathrm{Pic}^0(F) \cong \mathrm{Alb}(F)^\vee.$$

Consequently $\dim \mathrm{Pic}^0(F) = q(F)$.

Proof. The first Chern class identifies topologically trivial line bundles with unitary characters of $\pi_1(F)$ modulo periods. By Hodge theory, those characters are parametrized by harmonic $(0, 1)$ -forms modulo integral periods, hence by the dual torus to $\mathrm{Alb}(F)$. The dimensions agree because $\dim \mathrm{Alb}(F) = h^{1,0}(F) = q(F)$. $\square$

Remark 2.3. *The analytic message of Lecture 2 is not merely that $\mathrm{Pic}^0(F)$ exists. It is that the non-linear deformation problem for divisor classes on F has a tangent space that is visibly linear:*

$$T_0 \mathrm{Pic}^0(F) \cong H^1(F, \mathcal{O}_F).$$

This principle returns repeatedly in the algebraic theory.

3. Lectures 3–4 and the appendix: pre-schemes, functor of points, representability, and tangent spaces

3.1. The functor of points

For a scheme X define its functor of points

$$h_X : (\mathrm{Sch})^{\mathrm{op}} \rightarrow (\mathrm{Sets}), \quad h_X(T) = \mathrm{Hom}(T, X).$$

Theorem 3.1 (Yoneda, concrete form). *The assignment $X \mapsto h_X$ is fully faithful. Equivalently, for schemes X, Y there is a canonical bijection*

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Nat}(h_X, h_Y).$$

Proof. Given $f : X \rightarrow Y$, composition yields a natural transformation $h_X \rightarrow h_Y$. Conversely, let $\eta : h_X \rightarrow h_Y$ be natural. Evaluate at $T = X$ and at the identity map $\text{id}_X \in h_X(X)$; this gives a morphism $f := \eta_X(\text{id}_X) : X \rightarrow Y$. For any $u : T \rightarrow X$, naturality applied to u shows

$$\eta_T(u) = h_Y(u)(\eta_X(\text{id}_X)) = f \circ u.$$

Thus η is determined by f , proving bijectivity. $\square$

3.2. Representable functors and fibre products

A contravariant functor $F : (\text{Sch}/S)^{\text{op}} \rightarrow (\text{Sets})$ is *representable* if $F \cong h_X$ for some S -scheme X . Once one speaks this language, many geometric constructions become tautological.

Proposition 3.2. *Let X, Y, Z be S -schemes. Then the fibre product $X \times_Z Y$ represents the functor*

$$T \longmapsto h_X(T) \times_{h_Z(T)} h_Y(T).$$

Proof. For every T , by the universal property of fibre products, giving a morphism $T \rightarrow X \times_Z Y$ is equivalent to giving a pair of morphisms $T \rightarrow X, T \rightarrow Y$ with equal composites to Z . This is precisely the fibre product of the sets of T -points. $\square$

3.3. Zariski tangent spaces and dual numbers

Let X be a k -scheme and $x \in X(k)$. Put $D_k = \text{Spec } k[\varepsilon]/(\varepsilon^2)$.

Definition 3.3. *The Zariski tangent space at x is*

$$T_x X := \text{Hom}_x(D_k, X),$$

namely morphisms sending the closed point of D_k to x .

Proposition 3.4. *Let $x \in X(k)$ and $\mathfrak{m}_x \subset \mathcal{O}_{X,x}$ be the maximal ideal. There is a canonical k -vector-space isomorphism*

$$T_x X \cong \text{Hom}_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k).$$

Proof. A morphism $D_k \rightarrow X$ centered at x is equivalent to a local homomorphism

$$\mathcal{O}_{X,x} \longrightarrow k[\varepsilon]/(\varepsilon^2)$$

whose composition with $k[\varepsilon]/(\varepsilon^2) \rightarrow k$ is the residue map at x . Such a map is determined by its effect on $\mathfrak{m}_x$, and the condition $\varepsilon^2 = 0$ forces $\mathfrak{m}_x^2$ to map to zero. Therefore the map factors uniquely through a k -linear map $\mathfrak{m}_x/\mathfrak{m}_x^2 \rightarrow k$. The construction is reversible. $\square$

Remark 3.5. *When a functor F is representable by X , the tangent space to F at a k -point $x \in F(k)$ is automatically the tangent space of X at the corresponding point. This simple remark lies behind all deformation-theoretic dimension computations later on.*

4. Lecture 5: Proj and invertible sheaves

4.1. The Proj construction

Let $S = \bigoplus_{d \geq 0} S_d$ be a positively graded ring generated by S_1 over S_0 . Then $X = \text{Proj } S$ is covered by the standard opens

$$D_+(f) = \text{Spec}(S_{(f)})_0, \quad f \in S_+ \text{ homogeneous.}$$

For a graded S -module M one writes $\tilde{M}$ for the associated quasi-coherent sheaf on X and $M(n)$ for the degree shift.

Proposition 4.1. *The sheaf $\mathcal{O}_X(1) = \widetilde{S(1)}$ is invertible. More generally, each $\mathcal{O}_X(n)$ is invertible, and on the open set $D_+(f)$ the sheaf $\mathcal{O}_X(n)$ is generated by the symbol f^{-n} .*

Proof. On $D_+(f)$ the graded localization $S(1)_{(f)}$ is free of rank one over $(S_{(f)})_0$, generated by f^{-1} . Hence $\widetilde{S(1)}$ restricts to a free rank-one module on each standard open, which proves invertibility. The statement for $\mathcal{O}_X(n)$ follows by tensor powers. $\square$

4.2. Invertible sheaves and Čech cocycles

Let X be a scheme and $\mathfrak{U} = \{U_i\}$ an open cover.

Proposition 4.2. *Isomorphism classes of invertible sheaves on X are in bijection with Čech 1-cocycles with values in $\mathcal{O}_X^\times$ modulo coboundary. Consequently,*

$$\mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^\times).$$

Proof. Given an invertible sheaf $\mathcal{L}$, choose trivializations $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$. On overlaps one gets transition functions $g_{ij} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$ satisfying $g_{ij}g_{jk}g_{ki} = 1$. Changing trivializations multiplies the cocycle by a coboundary. Conversely, a cocycle glues the trivial bundles $\mathcal{O}_{U_i}$ into an invertible sheaf. The two constructions are inverse to one another. $\square$

4.3. The Picard group of projective space

Theorem 4.3. *Let k be an algebraically closed field. Then*

$$\mathrm{Pic}(\mathbf{P}_k^n) \cong \mathbf{Z},$$

generated by $\mathcal{O}_{\mathbf{P}^n}(1)$.

Proof. Every codimension-one integral subscheme of $\mathbf{P}^n$ is a hypersurface cut out by a single homogeneous irreducible polynomial, because the homogeneous coordinate ring is a UFD. Hence the divisor class group is generated by hyperplanes. Since $\mathbf{P}^n$ is smooth, Cartier and Weil divisor classes coincide; thus every line bundle is of the form $\mathcal{O}_{\mathbf{P}^n}(m)$ for a unique $m \in \mathbf{Z}$. $\square$

5. Lecture 6: properties of morphisms and coherent sheaves

5.1. Finite, proper, and projective morphisms

A morphism $f : X \rightarrow Y$ is *finite* if affine-locally it comes from a finite ring map; it is *projective* if it factors as a closed immersion into some $\mathbf{P}_Y^n$ followed by the projection; it is *proper* if it is separated, of finite type, and universally closed. Every projective morphism is proper.

The properties needed later are not the full EGA package, but the following two principles.

Proposition 5.1. *Let $f : X \rightarrow Y$ be finite.*

- (i) *For every coherent $\mathcal{F}$ on X , the direct image $f_*\mathcal{F}$ is coherent on Y .*
- (ii) *The scheme-theoretic fibre X_y over a point $y \in Y$ is finite, hence zero-dimensional.*

Proof. Both assertions are affine-local on Y . Write $Y = \mathrm{Spec} A$, $X = \mathrm{Spec} B$ with B finite over A . A coherent sheaf on X corresponds to a finite B -module M , which is automatically a finite A -module; this proves (i). For (ii), the fibre is $\mathrm{Spec}(B \otimes_A \kappa(y))$, a finite-dimensional $\kappa(y)$ -algebra, hence an Artinian scheme. $\square$

5.2. Associated points and purity issues

For a coherent sheaf $\mathcal{F}$ on a noetherian scheme X , the set $\text{Ass}(\mathcal{F})$ of associated points controls where nonzerodivisors live. The following criterion is used repeatedly when discussing Cartier divisors and flatness.

Lemma 5.2. *Let X be noetherian and $f \in \Gamma(X, \mathcal{O}_X)$. The multiplication map $\mathcal{F} \xrightarrow{f} \mathcal{F}$ is injective if and only if f avoids all associated points of $\mathcal{F}$.*

Proof. The statement is local. For a finite module M over a noetherian ring A , multiplication by f is injective precisely when f is not contained in any associated prime of M . This is the standard criterion for nonzerodivisors on finite modules. $\square$

5.3. Projective pushforward and eventual generation

Let $\pi : \mathbf{P}_S^n \rightarrow S$ be the projection. The deep theorems of coherent direct image, base change, and flattening are eventually proved by combining the explicit cohomology of twists on projective space with noetherian induction. We start with the explicit part in the next section.

6. Lecture 7: cohomology of coherent sheaves on projective space

6.1. The cohomology of $\mathcal{O}_{\mathbf{P}^n}(d)$

Let k be a field and $\mathbf{P}^n = \mathbf{P}_k^n$.

Theorem 6.1. *For every integer d one has:*

- (i) $H^0(\mathbf{P}^n, \mathcal{O}(d)) \cong \text{Sym}^d(k^{n+1})^\vee$ for $d \geq 0$, and $H^0(\mathbf{P}^n, \mathcal{O}(d)) = 0$ for $d < 0$.
- (ii) $H^i(\mathbf{P}^n, \mathcal{O}(d)) = 0$ for $0 < i < n$ and all d .
- (iii) By Serre duality,

$$H^n(\mathbf{P}^n, \mathcal{O}(d)) \cong H^0(\mathbf{P}^n, \mathcal{O}(-d - n - 1))^\vee,$$

so $H^n(\mathbf{P}^n, \mathcal{O}(d)) = 0$ for $d > -n - 1$.

Proof. Part (i) is immediate from the graded description of sections. For the vanishing, argue by induction on n . The case $n = 0$ is tautological. For $n \geq 1$, let $H \subset \mathbf{P}^n$ be a hyperplane. The exact sequence

$$0 \rightarrow \mathcal{O}_{\mathbf{P}^n}(d-1) \rightarrow \mathcal{O}_{\mathbf{P}^n}(d) \rightarrow \mathcal{O}_H(d) \rightarrow 0$$

yields long exact sequences in cohomology. By induction on n , the middle cohomology of $\mathcal{O}_H(d)$ vanishes for $0 < i < n - 1$. A descending induction on d then shows $H^i(\mathbf{P}^n, \mathcal{O}(d)) = 0$ for $0 < i < n$. Finally, the canonical bundle of $\mathbf{P}^n$ is $\mathcal{O}(-n - 1)$, and Serre duality gives (iii). $\square$

6.2. A relative version via Leray spectral sequences

Let S be a noetherian scheme and $\pi : \mathbf{P}_S^n \rightarrow S$ the projection.

Theorem 6.2. *For every integer d :*

- (i) If $d \geq 0$, then $\pi_* \mathcal{O}(d) \cong \text{Sym}^d(\mathcal{O}_S^{n+1})^\vee$ and $R^i \pi_* \mathcal{O}(d) = 0$ for $i > 0$.
- (ii) If $-n \leq d \leq -1$, then $R^i \pi_* \mathcal{O}(d) = 0$ for all i .
- (iii) If $d \leq -n - 1$, then $R^n \pi_* \mathcal{O}(d) \cong \text{Sym}^{-d-n-1}(\mathcal{O}_S^{n+1}) \otimes \det(\mathcal{O}_S^{n+1})^{-1}$ and $R^i \pi_* \mathcal{O}(d) = 0$ for $i \neq n$.

Proof. By cohomology-and-base-change, it suffices to compute the fibres at all geometric points of S , and those fibres are exactly the cohomology groups of [theorem 6.1](#). Since the fibre dimensions are constant in the displayed ranges, the corresponding higher direct images are locally free and commute with base change. The formula for $R^n \pi_* \mathcal{O}(d)$ follows from relative Serre duality. $\square$

Remark 6.3. *The Leray spectral sequence*

$$E_2^{p,q} = H^p(S, R^q \pi_* \mathcal{F}) \implies H^{p+q}(\mathbf{P}_S^n, \mathcal{F})$$

becomes effective only after the groups $R^q \pi_* \mathcal{F}$ are understood. For twists of the structure sheaf this explicit calculation is the basic engine behind the rest of the theory.

7. Lecture 8: flattening stratifications

7.1. Flatness and constant Hilbert polynomial

Let $\pi : \mathbf{P}_S^n \rightarrow S$ and let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}_S^n$. For a point $s \in S$ define the fibre Hilbert polynomial by

$$P_s(m) = \chi(\mathbf{P}_{\kappa(s)}^n, \mathcal{F}_s(m)).$$

If $\mathcal{F}$ is flat over S , then P_s is locally constant in s .

Proposition 7.1. *Assume $\mathcal{F}$ is flat over S . Then the function $s \mapsto P_s$ is locally constant on S .*

Proof. Choose $m \gg 0$ such that all fibres $\mathcal{F}_s$ are m -regular; this will be proved later using Castelnuovo–Mumford regularity. Then for every $r \geq m$, one has $H^i(\mathcal{F}_s(r)) = 0$ for $i > 0$, so

$$P_s(r) = h^0(\mathbf{P}_{\kappa(s)}^n, \mathcal{F}_s(r)).$$

By cohomology and base change, $\pi_* \mathcal{F}(r)$ is locally free and its rank equals $P_s(r)$, hence is locally constant. Since a polynomial of degree at most n is determined by finitely many values, P_s is locally constant. $\square$

7.2. Flattening strata via Fitting ideals

The deeper converse is that constant Hilbert polynomial can be imposed functorially by a universal stratification.

Theorem 7.2 (projective flattening stratification). *Let S be noetherian and $\mathcal{F}$ a coherent sheaf on $\mathbf{P}_S^n$. There exists a finite disjoint decomposition*

$$S = \coprod_P S_P$$

into locally closed subschemes indexed by polynomials $P \in \mathbf{Q}[m]$ such that for every morphism $T \rightarrow S$ the pullback $\mathcal{F}_T$ is flat over T with Hilbert polynomial P on every fibre if and only if $T \rightarrow S$ factors through S_P .

Proof in the projective case. Choose m so large that for every fibre $\mathcal{F}_s$ the sheaf $\mathcal{F}_s$ is m -regular. Then for $r = m, \dots, m+n$ the sheaves

$$E_r := \pi_* \mathcal{F}(r)$$

control the Hilbert polynomial, because on each fibre the higher cohomology vanishes and $\text{rk}(E_r \otimes \kappa(s)) = P_s(r)$. For each sequence of integers $(\rho_m, \dots, \rho_{m+n})$, the locus on which

all E_r have rank ρ_r is locally closed, cut out by Fitting ideals of the E_r . On that locus the multiplication maps by the homogeneous coordinates induce a graded module presentation of the fibre with the desired Hilbert function in degrees $m, \dots, m+n$; by regularity this determines the whole Hilbert polynomial. The universal property follows because flatness in this setting is equivalent to local freeness of these controlling direct images together with fibrewise vanishing of higher cohomology. $\square$

Remark 7.3. *Historically, Lecture 8 is the pivot between cohomological estimates and parameter spaces. Once flattening exists, one can carve out from a large Grassmannian or Quot-type ambient space the locus representing families of curves with fixed Hilbert polynomial.*

8. Lecture 9: Cartier divisors

8.1. The sheaf of total quotient rings

Let X be a noetherian scheme. There is a sheaf $\mathcal{K}_X$ characterized by the property that on every affine open $U = \text{Spec } A$,

$$\Gamma(U, \mathcal{K}_X) = Q(A),$$

where $Q(A)$ is the total quotient ring obtained by inverting the nonzerodivisors of A .

Definition 8.1. *A Cartier divisor on X is a global section of the quotient sheaf $\mathcal{K}_X^\times / \mathcal{O}_X^\times$. Thus a Cartier divisor may be represented by local equations $f_i \in \Gamma(U_i, \mathcal{K}_X^\times)$ such that $f_i/f_j \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times)$.*

Proposition 8.2. *There is an exact sequence of sheaves of abelian groups*

$$1 \rightarrow \mathcal{O}_X^\times \rightarrow \mathcal{K}_X^\times \rightarrow \text{Div}_X \rightarrow 0,$$

where Div_X denotes the sheaf of Cartier divisors.

Proof. By definition, $\text{Div}_X = \mathcal{K}_X^\times / \mathcal{O}_X^\times$, so exactness at the last term is tautological. Exactness at the middle term says precisely that a meromorphic function defines the trivial Cartier divisor iff it is everywhere a unit. $\square$

8.2. From divisors to invertible sheaves

Given a Cartier divisor D represented by local equations f_i , define the fractional ideal sheaf

$$\mathcal{O}_X(D)|_{U_i} = f_i^{-1} \mathcal{O}_{U_i} \subset \mathcal{K}_X.$$

Proposition 8.3. *The sheaf $\mathcal{O}_X(D)$ is invertible, and the assignment $D \mapsto \mathcal{O}_X(D)$ factors through linear equivalence to give a homomorphism*

$$\text{Cl}_{\text{Cart}}(X) \longrightarrow \text{Pic}(X).$$

If X is integral, regular, and noetherian, this map is an isomorphism.

Proof. On U_i the sheaf $\mathcal{O}_X(D)$ is generated inside $\mathcal{K}_X$ by the single element f_i^{-1} , hence is locally free of rank one. If $D = \text{div}(g)$ is principal, then multiplication by g^{-1} trivializes $\mathcal{O}_X(D)$ globally, so the assignment factors through divisor classes. On a regular integral scheme, every codimension-one prime is locally principal, hence Cartier, and every invertible sheaf is represented by a Cartier divisor after choosing a rational section. $\square$

8.3. Effective Cartier divisors

A Cartier divisor D is *effective* if it can be represented by local equations in $\mathcal{O}_X$. Equivalently, the inclusion $\mathcal{O}_X(-D) \subset \mathcal{O}_X$ defines a closed subscheme.

Lemma 8.4. *If X is regular and integral, then effective Cartier divisors are exactly closed subschemes of pure codimension one.*

Proof. If D is effective, its ideal is locally principal generated by a nonzerodivisor, so the quotient has codimension one and no embedded components. Conversely, on a regular scheme of dimension two, every height-one prime is principal in the local ring, hence any pure codimension-one closed subscheme is locally cut out by a nonzerodivisor. $\square$

9. Lecture 10: functorial properties of effective Cartier divisors

9.1. Pullback

Let $f : Y \rightarrow X$ and let $D \subset X$ be an effective Cartier divisor defined by an ideal sheaf $\mathcal{I}_D \subset \mathcal{O}_X$.

Definition 9.1. *Assume that $f^{-1}(D)$ contains no associated point of Y . Then the pullback divisor f^*D is the effective Cartier divisor on Y defined by the ideal $\mathcal{I}_D \cdot \mathcal{O}_Y$.*

Proposition 9.2. *Under the above hypothesis, one has a canonical isomorphism*

$$\mathcal{O}_Y(f^*D) \cong f^*\mathcal{O}_X(D).$$

Moreover pullback is compatible with composition.

Proof. Locally D is given by a nonzerodivisor $a \in A$, with $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$. The hypothesis precisely says that the image of a in B is a nonzerodivisor, so $(a) \subset B$ defines an effective Cartier divisor. The associated invertible sheaf is generated by a^{-1} inside the total quotient ring of B , which is exactly the pullback of the sheaf generated by a^{-1} on X . $\square$

9.2. Norms and pushforward along finite flat maps

Let $f : X \rightarrow Y$ be finite locally free of rank r . For a section $u \in \Gamma(X, \mathcal{O}_X)$, define the norm

$$\operatorname{Nm}_{X/Y}(u) = \det(m_u : f_*\mathcal{O}_X \rightarrow f_*\mathcal{O}_X).$$

Proposition 9.3. *Let $D \subset X$ be an effective Cartier divisor. Then there exists a unique effective Cartier divisor $\operatorname{Nm}_{X/Y}(D)$ on Y such that locally, if D is given by a nonzerodivisor u , then $\operatorname{Nm}_{X/Y}(D)$ is defined by $\operatorname{Nm}_{X/Y}(u)$. One has*

$$\mathcal{O}_Y(\operatorname{Nm}_{X/Y}(D)) \cong \operatorname{Nm}_{X/Y}(\mathcal{O}_X(D)).$$

Proof. Work affine-locally with $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$, where B is finite projective over A . If $u \in B$ is a nonzerodivisor, multiplication by u on B is injective, hence its determinant is a nonzerodivisor in A . Therefore $\operatorname{Nm}(u)$ defines an effective Cartier divisor on Y . Replacing u by vu with $v \in B^\times$ multiplies the norm by the unit $\operatorname{Nm}(v)$, so the resulting divisor is independent of the local equation. The statement on invertible sheaves follows from the determinant definition of the norm line bundle. $\square$

Remark 9.4. *This norm construction is the algebraic avatar of the pushforward of divisors under finite morphisms. On smooth curves it reduces to the usual sum over images weighted by residue-field degrees.*

10. Lecture 11: back to the classical case

Return to a smooth projective surface F/k over an algebraically closed field.

10.1. Riemann–Roch on curves and surfaces

For a smooth projective integral curve C and an invertible sheaf L on C ,

$$\chi(C, L) = \deg L + 1 - g(C).$$

On surfaces the fundamental formula is

$$\chi(F, \mathcal{O}_F(D)) = \frac{1}{2} D \cdot (D - K_F) + \chi(F, \mathcal{O}_F),$$

where K_F is a canonical divisor.

Remark 10.1. *In these notes we take the surface formula as standard input, whether from Grothendieck–Riemann–Roch or from a classical proof. What matters for the later lectures is how it specializes to curves lying on F .*

10.2. Adjunction and arithmetic genus

Let $C \subset F$ be an effective Cartier divisor. Then the conormal sequence yields

$$0 \rightarrow \mathcal{O}_F(-C)|_C \rightarrow \Omega_F^1|_C \rightarrow \Omega_C^1 \rightarrow 0,$$

from which one deduces the adjunction isomorphism

$$\omega_C \cong (\omega_F \otimes \mathcal{O}_F(C))|_C.$$

Proposition 10.2 (adjunction formula). *For an integral curve $C \subset F$ one has*

$$2p_a(C) - 2 = C \cdot (C + K_F).$$

Proof. By adjunction,

$$\deg \omega_C = (K_F + C) \cdot C.$$

For an integral projective curve, $\deg \omega_C = 2p_a(C) - 2$. Combining the two equalities gives the result. $\square$

Corollary 10.3. *For an effective Cartier divisor $C \subset F$,*

$$\chi(\mathcal{O}_C) = -\frac{1}{2}(C^2 + K_F \cdot C).$$

Equivalently,

$$p_a(C) = 1 + \frac{1}{2}(C^2 + K_F \cdot C).$$

Proof. From the exact sequence

$$0 \rightarrow \mathcal{O}_F(-C) \rightarrow \mathcal{O}_F \rightarrow \mathcal{O}_C \rightarrow 0$$

one gets

$$\chi(\mathcal{O}_C) = \chi(\mathcal{O}_F) - \chi(\mathcal{O}_F(-C)).$$

Apply Riemann–Roch on F to $-C$. $\square$

11. Lecture 12: the over-all classification of curves on surfaces

11.1. Hilbert polynomials and divisor classes

Let H be a fixed very ample divisor on F . For an effective divisor C the Hilbert polynomial is

$$P_C(m) = (H \cdot C)m - \frac{1}{2}(C^2 + K_F \cdot C).$$

Thus on a surface, fixing the Hilbert polynomial fixes two numerical invariants:

$$H \cdot C \quad \text{and} \quad p_a(C).$$

In favourable situations it fixes the numerical equivalence class of C outright.

11.2. The Hilbert functor of curves on F

For a polynomial $P \in \mathbf{Q}[m]$ define a functor on k -schemes:

$$\mathcal{Hilb}_F^P(T) = \left\{ \begin{array}{l} \text{closed subschemes } Z \subset F \times T \\ \text{flat over } T \text{ with } P_{Z_t} = P \end{array} \right\} / \cong.$$

Theorem 11.1 (Hilbert representability, surface form). *For every polynomial P , the functor $\mathcal{Hilb}_F^P$ is represented by a projective scheme Hilb_F^P .*

Proof sketch emphasizing the geometry. Choose an embedding $F \hookrightarrow \mathbf{P}^N$. For $m \gg 0$, every ideal sheaf $\mathcal{I}_Z$ with Hilbert polynomial P is m -regular, so $\mathcal{I}_Z(m)$ is generated by its global sections and has no higher cohomology. Therefore each such Z yields a quotient

$$H^0(F, \mathcal{I}_Z(m)) \otimes \mathcal{O}_F(-m) \rightarrow \mathcal{I}_Z,$$

with source of fixed rank determined by P . Hence $\mathcal{Hilb}_F^P$ embeds into a suitable Quot scheme, itself projective by Grothendieck. The flattening stratification from Lecture 8 cuts out exactly the locus of quotients corresponding to flat families of subschemes with Hilbert polynomial P . This locus represents the Hilbert functor. $\square$

11.3. The divisor locus

Inside Hilb_F^P there is an open subscheme Div_F^P parametrizing those subschemes which are effective Cartier divisors.

Proposition 11.2. *The locus $\text{Div}_F^P \subset \text{Hilb}_F^P$ of effective Cartier divisors is open.*

Proof. Being an effective Cartier divisor is equivalent, for a flat family of codimension-one subschemes $Z \subset F \times T$, to the condition that the ideal sheaf $\mathcal{I}_Z$ be invertible. For a coherent sheaf flat over the base, local freeness of rank one is an open condition on the base. Therefore the Cartier locus is open. $\square$

Remark 11.3. *This gives a modern incarnation of Mumford's spaces $C(t)$ and $P(t)$: one fixes the relevant Hilbert polynomial and then passes either to the divisor locus inside the Hilbert scheme or to its image in the Picard scheme.*

12. Lecture 13: linear systems and examples

12.1. Complete linear systems

Let D be a divisor on F . The associated complete linear system is

$$|D| := \mathbf{P}H^0(F, \mathcal{O}_F(D)).$$

Its points correspond to effective divisors linearly equivalent to D .

Proposition 12.1. *If $H^0(F, \mathcal{O}_F(D)) \neq 0$, then $|D|$ is naturally identified with the fibre of the map*

$$\mathrm{Div}_F^P \longrightarrow \mathrm{Pic}(F), \quad C \longmapsto \mathcal{O}_F(C),$$

over the point $[\mathcal{O}_F(D)]$, where $P = P_D$.

Proof. A nonzero section $s \in H^0(F, \mathcal{O}_F(D))$ determines its divisor of zeros $\mathrm{div}(s) + D$, which is an effective divisor linearly equivalent to D . Two sections define the same divisor iff they differ by a scalar. Conversely every effective divisor linearly equivalent to D arises from a nonzero section of $\mathcal{O}_F(D)$. $\square$

12.2. Examples

Example 12.2 (Plane curves). On $F = \mathbf{P}^2$, a divisor of degree d is cut out by a homogeneous polynomial of degree d . Hence

$$|dH| \cong \mathbf{P}H^0(\mathbf{P}^2, \mathcal{O}(d)) \cong \mathbf{P}^{\binom{d+2}{2}-1}.$$

The arithmetic genus is

$$p_a(dH) = \frac{(d-1)(d-2)}{2}.$$

Example 12.3 (Bidegree on $\mathbf{P}^1 \times \mathbf{P}^1$). If $F = \mathbf{P}^1 \times \mathbf{P}^1$ and $D = aF_1 + bF_2$, then

$$H^0(F, \mathcal{O}(a, b)) \cong H^0(\mathbf{P}^1, \mathcal{O}(a)) \otimes H^0(\mathbf{P}^1, \mathcal{O}(b)),$$

so

$$|aF_1 + bF_2| \cong \mathbf{P}^{(a+1)(b+1)-1}.$$

Moreover $p_a(D) = (a-1)(b-1)$.

Example 12.4 (Blow-up of the plane at one point). Let $\pi : \widetilde{\mathbf{P}}^2 \rightarrow \mathbf{P}^2$ be the blow-up at a point, with exceptional divisor E and $H = \pi^* \mathcal{O}_{\mathbf{P}^2}(1)$. Then divisors are of the form $dH - mE$. The linear system $|dH - mE|$ corresponds to plane curves of degree d having multiplicity at least m at the center. This is the first setting where the parameter space of curves is no longer a single projective space but a more delicate locally closed locus inside one.

13. Lecture 14: some vanishing theorems

13.1. Serre vanishing

Theorem 13.1 (Serre vanishing). *Let X be projective over a noetherian ring, let $\mathcal{L}$ be ample, and let $\mathcal{F}$ be coherent on X . Then for all $i > 0$,*

$$H^i(X, \mathcal{F} \otimes \mathcal{L}^{\otimes m}) = 0 \quad \text{for } m \gg 0.$$

Moreover $\mathcal{F} \otimes \mathcal{L}^{\otimes m}$ is generated by global sections for $m \gg 0$.

Proof. Embed X into some projective space so that $\mathcal{L}$ becomes a power of $\mathcal{O}_X(1)$. Resolve $\mathcal{F}$ by a finite complex of direct sums of negative twists of $\mathcal{O}_{\mathbf{P}^N}$. The cohomology of those twists is known from [theorem 6.1](#). Twisting far enough kills the higher cohomology of each term of the resolution, hence of $\mathcal{F}(m)$. The global-generation statement follows because the evaluation map from finitely many global sections becomes surjective after sufficiently positive twisting. $\square$

13.2. Castelnuovo–Mumford regularity

Definition 13.2. A coherent sheaf $\mathcal{F}$ on $\mathbf{P}^N$ is m -regular if

$$H^i(\mathbf{P}^N, \mathcal{F}(m-i)) = 0 \quad \text{for all } i > 0.$$

Theorem 13.3 (Mumford regularity theorem). If $\mathcal{F}$ is m -regular on $\mathbf{P}^N$, then:

- (i) $\mathcal{F}$ is $(m+1)$ -regular;
- (ii) for every $r \geq m$, the sheaf $\mathcal{F}(r)$ is generated by global sections;
- (iii) the multiplication maps

$$H^0(\mathcal{F}(r)) \otimes H^0(\mathcal{O}(1)) \rightarrow H^0(\mathcal{F}(r+1))$$

are surjective for every $r \geq m$.

Proof. Intersect with a hyperplane H avoiding the associated points of $\mathcal{F}$. Then the exact sequence

$$0 \rightarrow \mathcal{F}(r-1) \rightarrow \mathcal{F}(r) \rightarrow \mathcal{F}_H(r) \rightarrow 0$$

and induction on the ambient dimension show that regularity descends to H and propagates upward in r . The surjectivity of multiplication maps is read from the long exact sequences. Global generation follows by induction on the dimension and the surjectivity of sections at each step. $\square$

Corollary 13.4. Let F be a smooth projective surface and H a very ample divisor. For every invertible sheaf $\mathcal{L}$ on F there exists m_0 such that for all $m \geq m_0$:

- (i) $H^i(F, \mathcal{L}(mH)) = 0$ for $i > 0$;
- (ii) $\mathcal{L}(mH)$ is globally generated;
- (iii) $\mathcal{L}(mH)$ is very ample.

Proof. Choose an embedding $F \hookrightarrow \mathbf{P}^N$ by H . Serre vanishing gives (i) and global generation. For very ampleness, once $\mathcal{L}(mH)$ and $\mathcal{L}(mH) \otimes \mathfrak{m}_x$ are globally generated uniformly in x , sections separate points and tangent directions. $\square$

14. Lecture 15: universal families of curves

14.1. The universal closed subscheme

Let P be a Hilbert polynomial and write $H = \text{Hilb}_F^P$. By representability there is a universal closed subscheme

$$\mathcal{C} \subset F \times H,$$

flat over H , whose fibre over a point $[C] \in H$ is the corresponding curve $C \subset F$.

Proposition 14.1. *The restriction of $\mathcal{C}$ to the open Cartier locus $\text{Div}_F^P \subset H$ is an effective Cartier divisor on $F \times \text{Div}_F^P$ flat over the base.*

Proof. By construction the fibres are codimension-one subschemes. On the Cartier locus the universal ideal sheaf is invertible, hence cuts out an effective Cartier divisor. Flatness is built into the universal property of the Hilbert scheme. $\square$

14.2. The tangent space and the local-to-global Ext spectral sequence

Let $[C] \in \text{Hilb}_F^P$ correspond to an effective Cartier divisor $C \subset F$. Then the tangent space to the Hilbert scheme at $[C]$ is controlled by deformations of the ideal sheaf $\mathcal{I}_C$.

Theorem 14.2. *If $C \subset F$ is an effective Cartier divisor, then there are canonical identifications*

$$T_{[C]} \text{Hilb}_F^P \cong \text{Hom}(\mathcal{I}_C, \mathcal{O}_C) \cong H^0(C, \mathcal{N}_{C/F}),$$

and the obstruction space is

$$\text{Ext}^1(\mathcal{I}_C, \mathcal{O}_C) \cong H^1(C, \mathcal{N}_{C/F}).$$

In particular, if $H^1(C, \mathcal{N}_{C/F}) = 0$, then Hilb_F^P is smooth at $[C]$ of dimension $h^0(C, \mathcal{N}_{C/F})$.

Proof. First-order embedded deformations of $C \subset F$ over $D_k = \text{Spec } k[\varepsilon]/(\varepsilon^2)$ correspond to liftings of the exact sequence

$$0 \rightarrow \mathcal{I}_C \rightarrow \mathcal{O}_F \rightarrow \mathcal{O}_C \rightarrow 0$$

to $F \times D_k$. Standard deformation theory identifies these with $\text{Hom}(\mathcal{I}_C, \mathcal{O}_C)$. Since C is Cartier, $\mathcal{I}_C \cong \mathcal{O}_F(-C)$ is invertible, so

$$\text{Hom}(\mathcal{I}_C, \mathcal{O}_C) \cong \mathcal{O}_C(C) = \mathcal{N}_{C/F}.$$

Now the local-to-global Ext spectral sequence

$$E_2^{p,q} = H^p(F, \mathcal{E}xt^q(\mathcal{I}_C, \mathcal{O}_C)) \implies \text{Ext}^{p+q}(\mathcal{I}_C, \mathcal{O}_C)$$

degenerates because $\mathcal{I}_C$ is locally free, hence $\mathcal{E}xt^q(\mathcal{I}_C, \mathcal{O}_C) = 0$ for $q > 0$. Therefore

$$\text{Ext}^i(\mathcal{I}_C, \mathcal{O}_C) \cong H^i(F, \mathcal{H}om(\mathcal{I}_C, \mathcal{O}_C)) \cong H^i(C, \mathcal{N}_{C/F})$$

for $i = 0, 1$. The smoothness criterion follows from standard obstruction theory. $\square$

14.3. The Abel map to the Picard scheme

On the divisor locus there is a natural morphism

$$\alpha : \text{Div}_F^P \longrightarrow \text{Pic}(F), \quad C \longmapsto \mathcal{O}_F(C).$$

Its fibres are complete linear systems. The tangent map at $[C]$ fits into the exact sequence derived from

$$0 \rightarrow \mathcal{O}_F \rightarrow \mathcal{O}_F(C) \rightarrow \mathcal{N}_{C/F} \rightarrow 0,$$

namely

$$0 \rightarrow H^0(F, \mathcal{O}_F) \rightarrow H^0(F, \mathcal{O}_F(C)) \rightarrow H^0(C, \mathcal{N}_{C/F}) \xrightarrow{d\alpha} H^1(F, \mathcal{O}_F).$$

Thus the differential of the Abel map is the cohomological bridge between linear systems and the irregularity of the surface.

15. Lecture 16: the method of Chow schemes

15.1. Chow forms of cycles in projective space

Let V be a vector space of dimension $N + 1$ and $\mathbf{P}(V) = \mathbf{P}^N$. For an effective cycle Z of pure dimension r and degree d in $\mathbf{P}(V)$, the *Chow form* is a multihomogeneous equation on $(\mathbf{P}(V^\vee))^{r+1}$ whose vanishing encodes the condition that $r + 1$ hyperplanes meet Z nontrivially. For curves ($r = 1$) in $\mathbf{P}^N$, one obtains a bihomogeneous form of degree d in each block of hyperplane coordinates.

Theorem 15.1 (existence of the Chow variety of curves in projective space). *For every $d \geq 0$ there exists a projective scheme $\text{Chow}_{1,d}(\mathbf{P}^N)$ representing effective one-cycles of degree d in $\mathbf{P}^N$ up to rational equivalence of coefficients, and its points correspond to Chow forms of degree d satisfying explicit incidence relations.*

Modern construction with classical interpretation. Classically, one embeds the set of Chow forms into a projective space of multihomogeneous forms and cuts it out by polynomial relations expressing multilinearity and the incidence condition with $(N - 2)$ -planes. Modernly, one may construct $\text{Chow}_{1,d}(\mathbf{P}^N)$ as the image of the Hilbert scheme of one-dimensional subschemes of degree d under the Hilbert–Chow morphism. Projectivity follows because the Hilbert scheme is projective and the morphism is proper. On geometric points this image identifies precisely with effective one-cycles of degree d , and the corresponding line in the coefficient space is the classical Chow form. $\square$

15.2. Curves on a fixed surface

Fix an embedding $F \hookrightarrow \mathbf{P}^N$. The Chow variety of one-cycles on F of degree d is the closed subvariety

$$\text{Chow}_{1,d}(F) \subset \text{Chow}_{1,d}(\mathbf{P}^N)$$

consisting of cycles supported on F .

Proposition 15.2. *There is a proper Hilbert–Chow morphism*

$$\rho : \text{Hilb}_F^P \longrightarrow \text{Chow}_{1,d}(F),$$

where d is the leading coefficient of P , and on the open locus of reduced Cartier divisors this map sends a curve to its underlying cycle with multiplicities.

Proof. The universal family on $F \times \text{Hilb}_F^P$ determines a relative fundamental cycle, hence a morphism to the Chow variety. Properness follows from properness of the Hilbert scheme. On a reduced Cartier divisor, the fundamental cycle is exactly the sum of irreducible components with their scheme-theoretic multiplicities. $\square$

15.3. Why Mumford’s route through Chow coordinates matters

The Hilbert scheme remembers embedded and nonreduced structure; the Chow scheme remembers only cycles. For families of divisors on a smooth surface, both viewpoints are useful:

- the Hilbert scheme is ideal for deformation theory and tangent-space calculations;
- the Chow scheme is ideal for degree bounds, compactness of cycle spaces, and elimination-theoretic constructions.

This is why Lecture 16 is not a historical detour but the second half of the parameter-space story.

Synthesis: from Lecture 1 to Lecture 16

The architecture can now be read as a single chain of implications.

- (1) Curves on a smooth surface are divisors, hence line bundles with sections.
- (2) Their numerical behaviour is encoded by intersection theory and Hilbert polynomials.
- (3) Their infinitesimal behaviour is encoded by tangent spaces and normal bundles.
- (4) Cohomology on projective space supplies the vanishing theorems and regularity needed to control families.
- (5) Flattening stratifications carve out loci of constant Hilbert polynomial.
- (6) The Hilbert scheme then provides a projective parameter space with universal family.
- (7) The divisor locus and the Picard map recover linear systems and the irregularity of the surface.
- (8) The Hilbert–Chow morphism reconnects the modern representability theorem with the classical Chow-form method.

That is precisely the conceptual content of Mumford’s first sixteen lectures, now rewritten in a scheme-theoretic and deformation-theoretic language suitable for contemporary use.

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